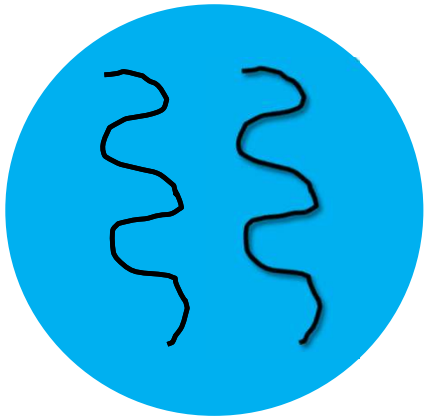
IMPLEMENTAREA CONCURENTEI IN LIMBAJE DE PROGRAMARE

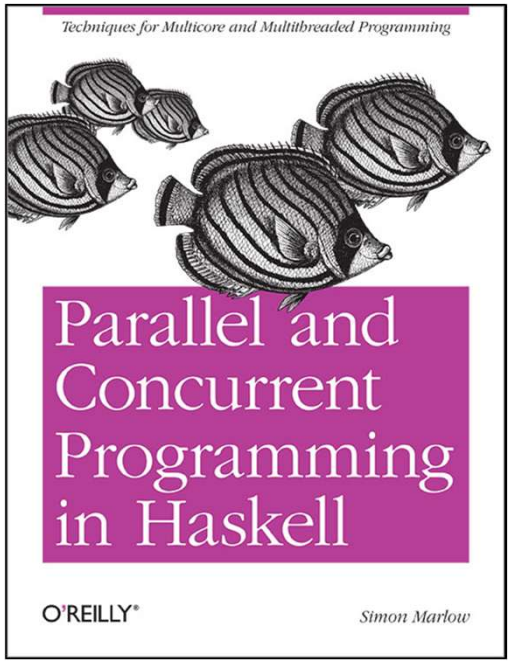
Concurenta

Threaduri

Memorie Partajata

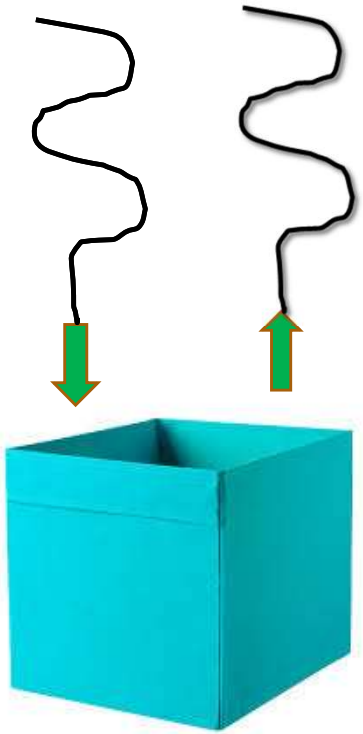
Ioana Leustean





Part II. Concurrent Haskell
Cap.7 & 8

https://hackage.haskell.org/package/base-4.20.0.1/docs/Control-Concurrent.html



forkIO :: IO () -> IO ThreadId

MVar
mutable variable

➢ data MVar a

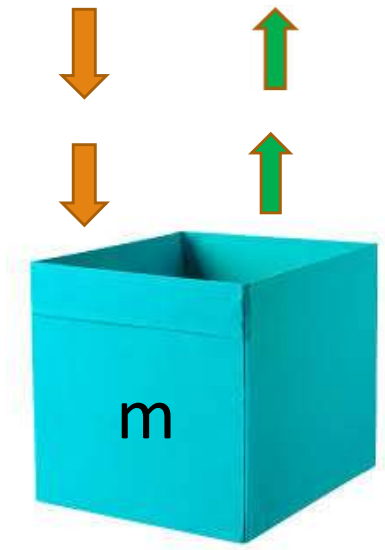
newEmptyMVar :: IO (MVar a) -- m <- newEmptyMVar
-- m este o locatie goala

newMVar :: a -> IO (MVar a) -- m <- newMVar v
-- m este o locatie care contine valoarea v

putMVar :: Mvar a -> a -> IO() -- putMVar m v
-- pune in m valoarea v
-- asteapta (blocheaza thread-ul) daca m este plina

takeMVar :: MVar a -> IO a -- v <- takeMVar m
-- intoarce in v valoarea din m si **goleste** m
-- asteapta (blocheaza thread-ul) daca m este goala

readMVar :: MVar a -> IO a -- v <- readMVar m citeste atomic continutul lui m
-- intoarce in v valoarea din m dar aceasta ramane in m
-- asteapta (blocheaza thread-ul) daca m este goala



➢ Semafor cu cantitate (quantity semaphore)

implementare existenta

```
import Control.Concurrent.QSem

data  QSem

newQSem :: Int -> IO  Qsem


waitQSem ::  QSem -> IO()        -- aquire, il ocupa
signalQSem :: QSem -> IO()       -- release, il elibereaza
```

un semafor care sincronizeaza accesul la n resurse se defineste astfel:

```
q <- newQsem  n
```

➢ Exemplu: **QSem**

O multime de taskuri acceseaza simultan o resursa reprezentata printr-un **QSem**;
pentru a se executa, fiecare task trebuie sa acceseaze resursa, pe care o elibereaza la sfarsitul executiei.

```
import Control.Concurrent
import Control.Monad


main :: IO ()
main = do
        q <- newQSem 3
        let workers = 5
        mapM_ (forkIO . worker q m) [1..workers]
```

mapM_ :: (Foldable t, Monad m) => (a -> m b) -> t a -> m ()

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main = do
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        let workers = 5
        mapM_ (forkIO . worker q m) [1..workers]
```

q este semaforul care controleaza resursele

```
worker ::  QSem -> MVar String -> Int -> IO ()
worker q m w= do
    waitQSem q
    putStrLn$ "Worker " ++ show w ++ " acquired the lock."
    threadDelay 2000000      -- microseconds
    signalQSem q
    putStrLn $ "Worker " ++ show w ++ "released the lock."
```

http://rosettacode.org/wiki/Metered_concurrency

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    y 2000000      -- microseconds
    signalQSem q
    putStrLn $ "Worker " ++ show w ++ "released the lock."
```

```
Ok, one module loaded.
*Main> main
WoWWo*Main> orrekkree rr1   23h  ahhsaa ssa  caaqccuqqiuuriierrdee ddt  httehh eel  ollcookcc.kk
..

WoWWrWWookoorrerrkkrkkee eerr1rr     23h54  a  hhshhaa aassrss  e  rrlaaeeeccllaqqeesuuaaeiissdrree eeddtdd  h
 ttetthh hheelee  o  llcllookoocc.cckk
kk....
```

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```
import Control.Concurrent
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main :: IO ()
main = do
        q <- newQSem 3
        let workers = 5
        mapM_ (forkIO . worker q ) [1..workers]
```

q este semaforul care controleaza resursele

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worker ::  QSem  -> Int -> IO ()
worker q m w= do
    waitQSem q
    putStrLn$ "Worker " ++ show w ++ " acquired the lock."
    y 2000000        -- microseconds
    signalQSem q
    putStrLn $ "Worker " ++ show w ++ "released the lock."
```

```
Ok, one module loaded.
*Main> main
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..

WoWWrWWookoorrerrkkrkkee eerr1rr     23h54  a  hhshhaa aassrss  e  rrlaaeeeccllaqqeesuuaaeiissdrree eeddtdd  h
 ttetthh hheelee  o  llcllookoocc.cckk
kk....
```

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➢ Exemplu: **QSem**

O multime de taskuri acceseaza simultan o resursa reprezentata printr-un **QSem**;
pentru a se executa, fiecare task trebuie sa acceseaze resursa, pe care o elibereaza la sfarsitul executiei.

```
import Control.Concurrent
import Control.Monad


main :: IO ()
main = do
        q <- newQSem 3
        stdo <- newEmptyMVar
        let workers = 5
            prints  = 2 * workers
        mapM_ (forkIO . worker q  stdo) [1..workers
        replicateM_ prints $ takeprint  stdo
```

```
takeprint :: MVar String  -> IO()
takeprint stdo = do
        s <- takeMVar stdo
        print s

worker ::  QSem -> MVar String -> Int -> IO ()
worker q stdo w= do
   waitQSem q
   putMVar stdo $ "Worker " ++ show w ++ " acquired the lock."
   threadDelay 2000000      -- microseconds
   signalQSem q
   putMVar stdo $ "Worker " ++ show w ++ "released the lock."
```

q este semaforul care controleaza resursele
stdo coordoneaza accesul la stdout

http://rosettacode.org/wiki/Metered_concurrency

```
Prelude> :l qsemrcmy.hs
[1 of 1] Compiling Main             ( qsemrcmy.hs, interpreted )
Ok, modules loaded: Main.
*Main> main
"Worker 1 has acquired the lock."
"Worker 2 has acquired the lock."
"Worker 3 has acquired the lock."
"Worker 2 has released the lock."
"Worker 3 has released the lock."
"Worker 1 has released the lock."
"Worker 5 has acquired the lock."
"Worker 4 has acquired the lock."
"Worker 4 has released the lock."
"Worker 5 has released the lock."
```

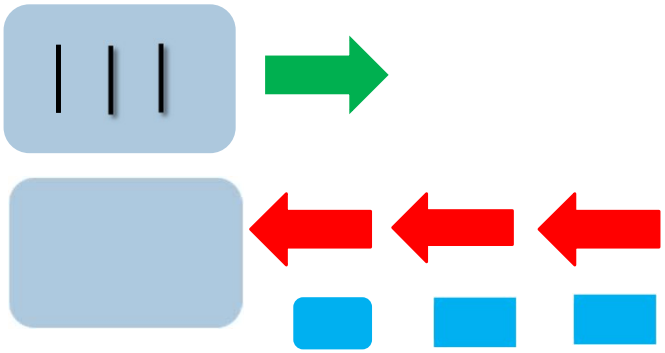
```
*Main> main
"Worker 1 has acquired the lock."
"Worker 2 has acquired the lock."
"Worker 3 has acquired the lock."
"Worker 1 has released the lock."
"Worker 5 has acquired the lock."
"Worker 2 has released the lock."
"Worker 4 has acquired the lock."
"Worker 3 has released the lock."
"Worker 4 has released the lock."
"Worker 5 has released the lock."
```

in Concurrent Haskell
concurenta este nedeterminista

➢ Implementarea QSem

```
type  QSem = MVar (?????)

waitQSem ::  QSem -> IO()          -- ocupa
signalQSem :: QSem -> IO()         -- elibereaza
```

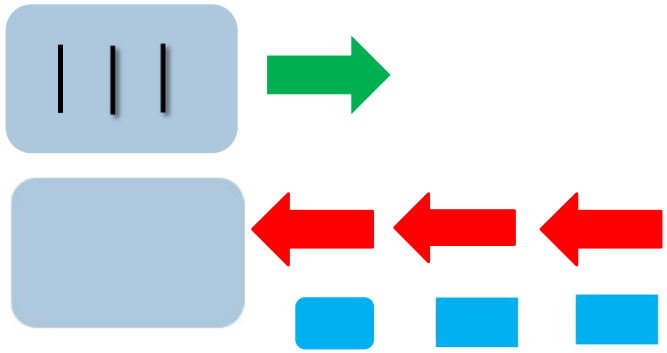


Trebuie sa reprezentam:

- numarul de resurse si mecanismul de blocare a thread-urilor
- daca numarul de resurse este n functionalitatea este dictata valoarea lui n:
  - n>0 (exista resurse): thread-urile care solicita accesul nu sunt blocate, dar micsoreaza numarul de resurse
  - n=0 (nu exista resurse): thread-urile care solicita accesul sunt blocate (in ordinea venirii) cand o resursa devine libera, primul thread care asteapta este deblocat

Implementarea din: *Concurrent Haskell, SL Peyton Jones, A Gordon, S Finne, 1996*

➢ Implementarea QSem



```
type  QSem = MVar (Int, [MVar ()])

newQSem :: Int -> IO  QSem

newQSem n =  newMVar (n,[])
                    --  qsem <- newQSem 3

waitQSem ::  QSem -> IO()          -- ocupa
signalQSem :: QSem -> IO()         -- elibereaza
```

n = nr. de resurse
blki = un thread care cere acces la resursa
este blocat pe variabila blki

daca n > 0 atunci qsem = (n, [])
altfel qsem = (0, [blk1, blk2, ...])

Implementarea din:
Concurrent Haskell
SL Peyton Jones, A Gordon, S Finne, 1996

- Implementarea QSem - *Concurrent Haskell SL Peyton Jones, A Gordon, S Finne, 1996*

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```

daca n > 0 atunci qsem = (n, [])
altfel qsem = (0, [blk1, blk2, …])

Ocuparea resursei

```
waitQSem ::  QSem -> IO()
waitQSem qsem = do
            (avail,blks) <- takeMVar qsem
            if avail > 0
                then    putMVar qsem (avail-1, [])
                else
                    do
                      blk <- newEmptyMVar
                      putMVar qsem (0, blk:blks)
                      takeMVar blk  -- threadul e blocat pe  variabila proprie
```

- Implementarea QSem - *Concurrent Haskell SL Peyton Jones, A Gordon, S Finne, 1996*

```
type  QSem = MVar (Int, [MVar ()])

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newQSem n =  newMVar (n,[])
```

daca n > 0 atunci qsem = (n, [])
altfel qsem = (0, [blk1, blk2, …])

Eliberarea resursei

```
signalQSem :: QSem -> IO()
signalQSem qsem = do
                    (avail,blks) <- takeMVar qsem
                    case blks of
                      [] -> putMVar qsem (avail+1,[])
                      (blk:blks') -> do
                              putMVar qsem (0,blks')
                              putMVar blk ()
```

- fiecare thread elibereaza variabila proprie a unui thread in asteptare

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Ocuparea resursei

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waitQSem ::  QSem -> IO()
waitQSem qsem = do
        (avail,blks) <- takeMVar qsem
        if avail > 0
            then    putMVar qsem (avail-1, [])
            else
                do
                  blk <- newEmptyMVar
                  putMVar qsem (0, blk:blks)
                  takeMVar blk
```

– threadul e blocat pe variabila proprie

atentie la ordinea de asteptare!

Eliberarea resursei

```
signalQSem :: QSem -> IO()
signalQSem qsem = do
        (avail,blks) <- takeMVar qsem
        case blks of
          [] -> putMVar qsem (avail+1,[])
          (blk:blks') -> do
                putMVar qsem (0,blks')
                putMVar blk ()
```

atentie la ordinea de asteptare!

fiecare thread elibereaza variabila proprie a unui thread in asteptare

➢ Cititori/scriitori (Readers/Writers)

- Mai multe threaduri au acces la o resursa.
- Unele threaduri scriu (writers), iar altele citesc (readers).
- Resursa poate fi accesata simultan de mai multi cititori.
- Resursa poate fi acessata de un singur scriitor.
- Resursa nu poate fi accesata simultan de cititori si de scriitori.

```
import  Control.Concurrent.ReadWriteLock
new  :: IO RWLock
acquireRead ::  IO RWLock -> IO ()
releaseRead ::  IO RWLock -> IO ()
acquireWrite ::  IO RWLock -> IO ()
releaseWrite ::  IO RWLock -> IO ()
```

➢ Cititori/scriitori (Readers/Writers)

Mai multe thread-uri au acces la o resursa.
Unele thread-uri scriu (writers), iar altele citesc (readers).
Resursa poate fi accesata simultan de mai multi cititori.
Resursa poate fi acessata de un singur scriitor.
Resursa nu poate fi accesata simultan de cititori si de scriitori.

Pentru **sincronizare** folosim:

- un semafor binar care da acces la citit sau la scris: writeL
- un monitor in care se inregistreaza nr. de cititori: readL

```
data MyRWLock = MyRWL {readL :: MVar Int, writeL :: MVar ()}
```

- Reader/Writer Lock

```
type MyLock = MVar ()
newLock = newMVar ()
aquireLock  m = takeMVar m
releaseLock m = putMVar m ()
```

```
data MyRWLock = MyRWL {readL :: MVar Int, writeL :: MyLock}
```

```
newMyRWLock :: IO MyRWLock
newMyRWLock = do
                readL  <-  newMVar 0
                writeL <-  newLock
                return (MyRWL readL writeL)
```

- Reader/Writer Lock

```
type MyLock = MVar ()
newLock = newMVar ()
aquireLock  m = takeMVar m
releaseLock m = putMVar m ()
```

```
data MyRWLock = MyRWL {readL :: MVar Int, writeL :: MyLock}
```

```
aquireWrite :: MyRWLock -> IO ()
aquireWrite (MyRWL readL  writeL) = aquireLock writeL

releaseWrite :: MyRWLock -> IO ()
releaseWrite (MyRWL readL writeL) = releaseLock writeL
```

- Reader/Writer Lock

```
data MyRWLock = MyRWL {readL :: MVar Int, writeL :: MyLock}
```

```
aquireRead :: MyRWLock -> IO ()
aquireRead (MyRWL readL writeL) = do
                    n <- takeMVar readL      -- n cititori
                    if (n == 0)   then do
                                  aquireLock writeL
                                  putMVar readL 1
                    else  putMVar readL (n+1)
```

- Reader/Writer Lock

```
data MyRWLock = MyRWL {readL :: MVar Int, writeL :: MyLock}
```

```
releaseRead :: MyRWLock -> IO ()
releaseRead (MyRWL readL writeL) = do
                                    n <- takeMVar readL
                                    if (n == 1)   then do
                                                  releaseLock writeL
                                                  putMVar readL 0
                                     else  putMVar readL (n-1)
```

➢ Exemplu: Readers/Writers

lib este resursa partajata
rwl este lacatul care sincronizeaza accesul la resursa

```
reader i rwl lib = do                    -- un thread cititor
                    aquireRead rwl
                    c <- readMVar lib  -- non blocking
                    putStrLn $  "Reader " ++ (show i) ++ " reads: " ++ (show c)
                    releaseRead rwl
```

➢ Exemplu: Readers/Writers

lib este resursa partajata
rwl este lacatul care sincronizeaza accesul la resursa

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reader i rwl lib = do                    -- un thread cititor
                    aquireRead rwl
                    c <- readMVar lib  -- non blocking
                    putStrLn $  "Reader " ++ (show i) ++ " reads: " ++ (show c)
                    releaseRead rwl


writer i rwl lib = do                    -- un thread scriitor
                     aquireWrite rwl
                     putStrLn $  "Writer " ++ (show i) ++ " writes "  (show i)
                      c <- takeMVar lib
                      putMVar lib i
                      releaseWrite rwl
```

- Exemplu: Readers/Writers

```
genread n rwl lib = if (n==0)
                    then putStrLn "no more readers"
                    else do
                        reader n rwl lib
                        threadDelay 20
                        genread (n-1) rwl lib
genwrite n rwl lib = if (n==0)
                     then putStrLn "no more writers"
                     else do
                        writer n rwl lib
                        threadDelay 100
                        genwrite (n-1) rwl lib

main = do
        lib <- newMVar 0      -- resursa
        rwl <- newMyRWLock    -- lacatul rw
        forkIO $  genread 10 rwl lib    -- creez 10 thread-uri cititor
        forkIO $  genwrite 5 rwl lib    -- creez 5 thread-uri scriitor
        getLine
```

```
reader i rwl lib = do
                    aquireRead rwl
                    c <- readMVar lib
                    putStrLn $  (show i) ++ (show c)
                    releaseRead rwl

writer i rwl lib = do
                    aquireWrite rwl
                    putStrLn $ show  i
                    c <- takeMVar lib
                    putMVar lib i
                    releaseWrite rwl
```

- Readers/Writers

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genread n rwl lib = if (n==0)
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                         threadDelay 20
                         genread (n-1) rwl lib
genwrite n rwl lib = if (n==0)
                     then putStrLn "no more writers"
                     else do
                         writer n rwl lib
                         threadDelay 100
                         genwrite (n-1) rwl lib

main = do
        lib <- newMVar 0
        rwl <- newMyRWLock
        forkIO $  genread 10 rwl lib
        forkIO $  genwrite 5 rwl lib
        getLine
```

```
reader i rwl lib = do
                    aquireRead rwl
                    c <- readMVar lib
                    putStrLn $  (show i) ++ (show c)
                    releaseRead rwl

writer i rwl lib = do
                    aquireWrite rwl
                    putStrLn $ show  i
                    c <- takeMVar lib
                    putMVar lib i
                    releaseWrite rwl
```

```
Prelude>  :l myrw.hs
[1 of 1] Compiling Main
Ok, modules loaded: Main.
*Main> main
Reader 10 reads: 0
Writer 5 writes 5
Reader 9 reads: 5
Reader 8 reads: 5
Writer 4 writes 4
Reader 7 reads: 4
Reader 6 reads: 4
Writer 3 writes 3
Reader 5 reads: 3
Writer 2 writes 2
Reader 4 reads: 2
Writer 1 writes 1
Reader 3 reads: 1
no more writers
Reader 2 reads: 1
Reader 1 reads: 1
no more readers
```

Canale de comunicare formate din variabile MVar

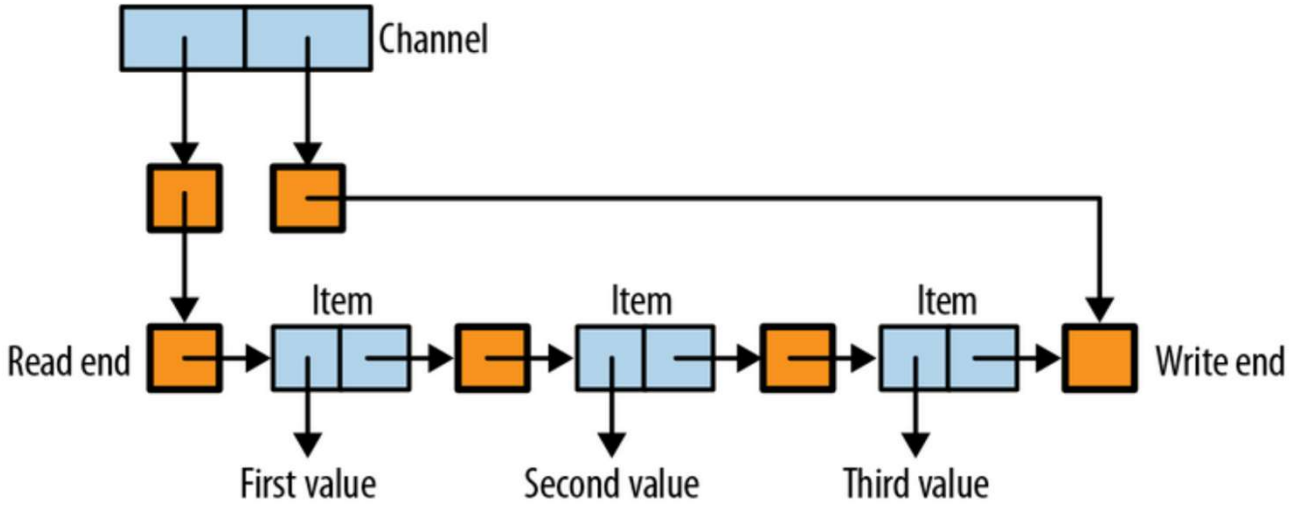
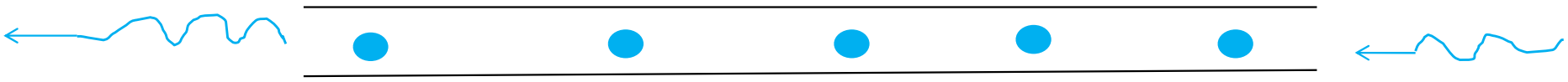


Figure 7-1. Structure of the buffered channel implementation

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Canale de comunicare: canale implementate cu MVar



readChan
se
blocheaza
cand
canalul
este gol

writeChan
nu se blocheaza
niciodata

```
import Control.Concurrent.Chan

newChan :: IO (Chan a)

writeChan :: Chan a -> a -> IO ()

readChan :: Chan  a -> IO a
```

➢ Exemplu: doua canale: **cin** si **cout**

- thread –ul parinte citeste siruri si le pune pe canalul **cin**.
- un thread citeste sirurile de pe **cin**, le imparte in cuvinte iar cuvintele le pune pe canalul **cout**.
- un alt thread ia cuvintele de pe **cout**, si le scrie la iesire cu litere mari

```
import Control.Monad
import Control.Concurrent
import Data.Char

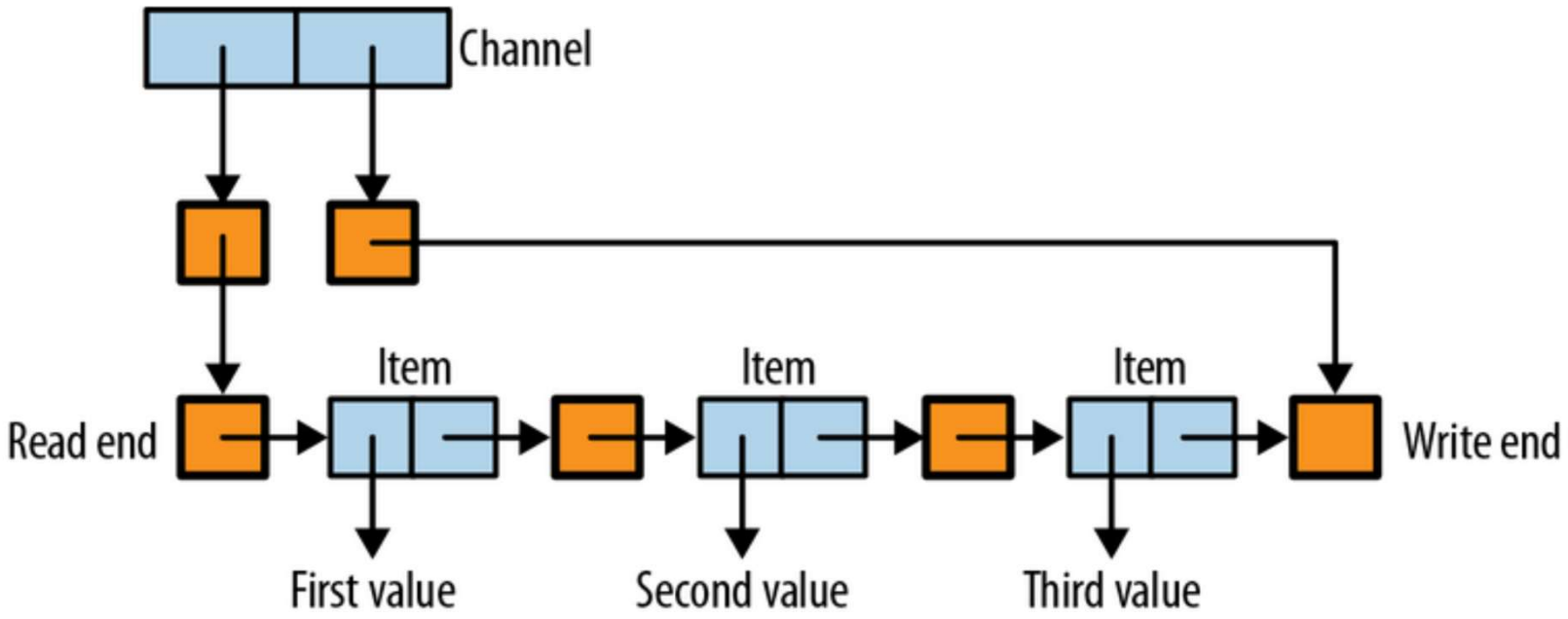
mymain = do
        cin  <- newChan
        cout <- newChan
       forkIO $ forever (move cin cout)
       forkIO $ forever (upout cout)
       load cin
```

```
move c1 c2 = do
              v1 <- readChan c1
              let ls = words v1
              mapM_ (writeChan c2) ls

upout c  = do
             str <- readChan c
             putStrLn (map toUpper str)

load c = do
             str <- getLine
             if (str == "exit")
             then return()
            else do
                   writeChan c str
                   load
```

- Canale de comunicare formate din variabile MVar



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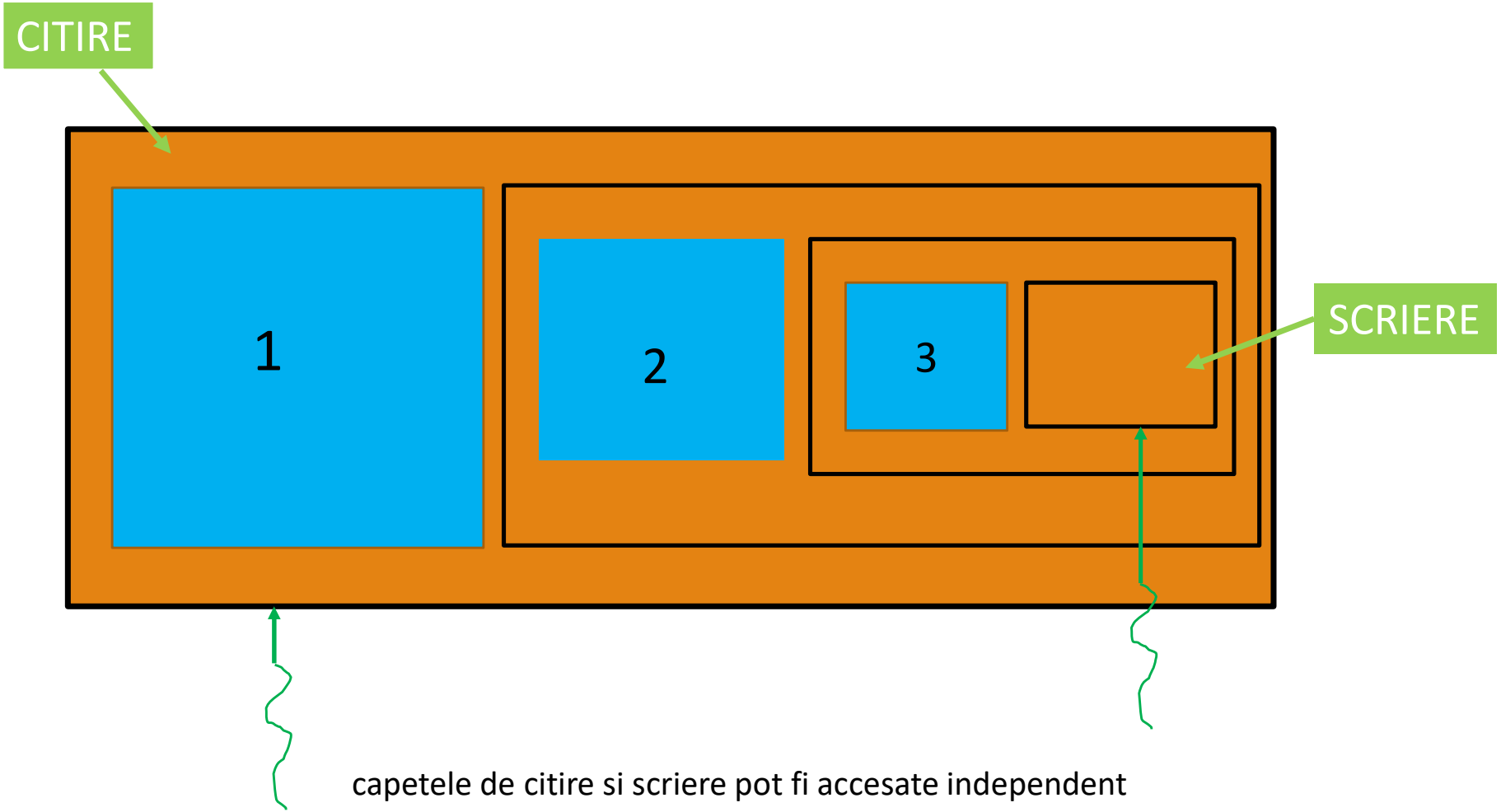


capetele de citire si scriere pot fi accesate independent

➢ Canale formate din variabile MVar

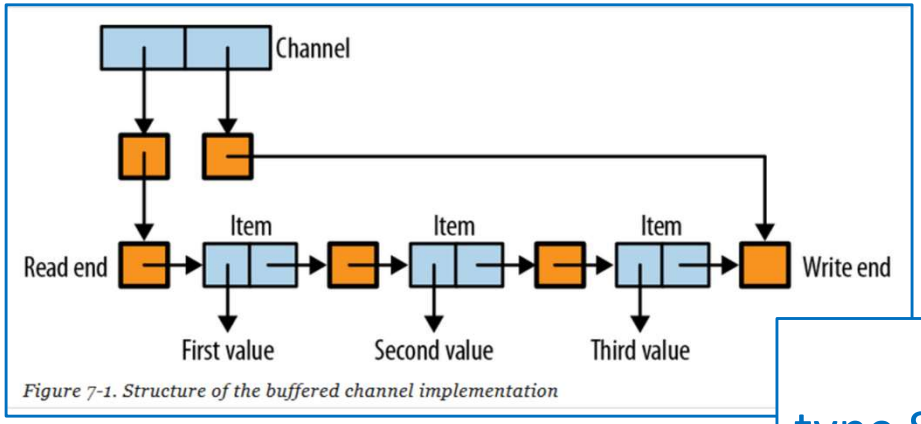
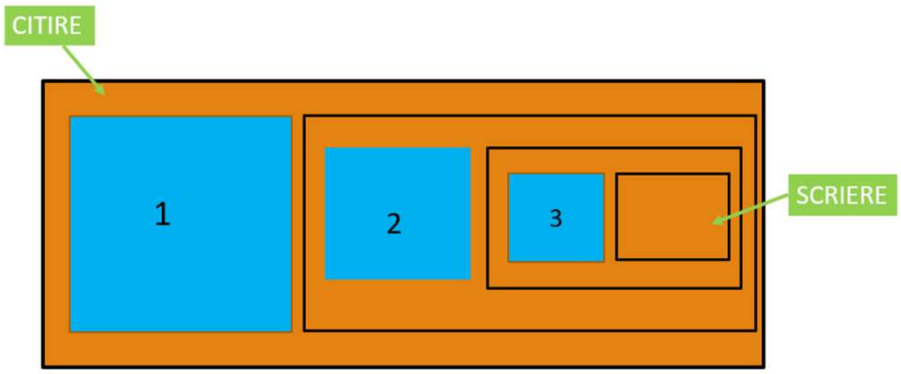


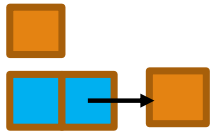
Figure 7-1. Structure of the buffered channel implementation




```
type Stream a = MVar (Item a)
data Item a    =  Item a (Stream a)

data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))

c <- newChan
v <- readChan c
putChan c v
```



chan.hs ©2012, Simon Marlow

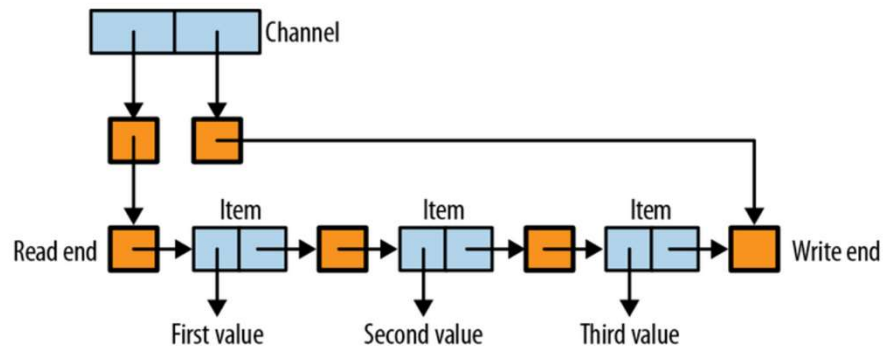


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))

c <- newChan
v <- readChan c
putChan c v
```

"If multiple threads concurrently call readChan, the first one will successfully call takeMVar on the read end, but the subsequent threads will all block at this point until the first thread completes the operation and updates the read end. If multiple threads call writeChan, a similar thing happens: the write end of the Chan is the synchronization point, allowing only one thread at a time to add an item to the channel. However, the read and write ends, being separate MVars, allow concurrent readChan and writeChan operations to proceed without interference."

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Implementarea canalelor

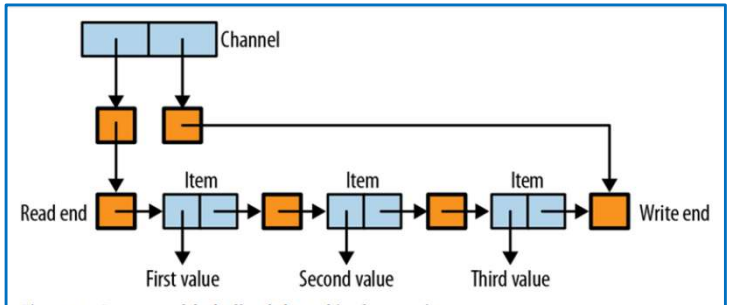


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
```

```
newChan :: IO(Chan a)
newChan = do
    emptyStream <-  newEmptyMVar
    readVar <- newMVar emptyStream
    writeVar <-newMVar emptyStream
    return (Chan readVar writeVar)
```

contine Item-ul care va fi citit

contine variabila in care se va scrie noul Item

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Implementarea canalelor

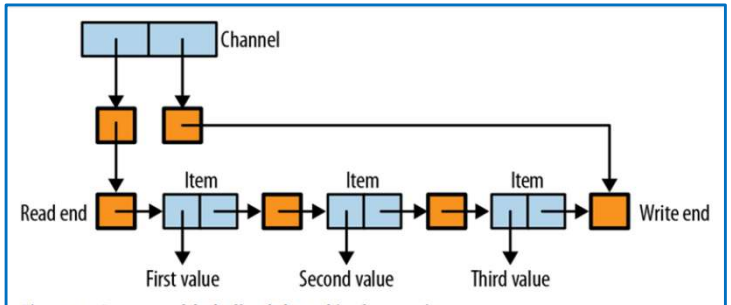


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
```

```
readChan :: Chan a -> IO a
readChan (Chan rV wV) = do
                    stream <- takeMVar rV
                    Item val str <- takeMVar stream
                    putMVar rV  str
                    return val
```

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Implementarea canalelor

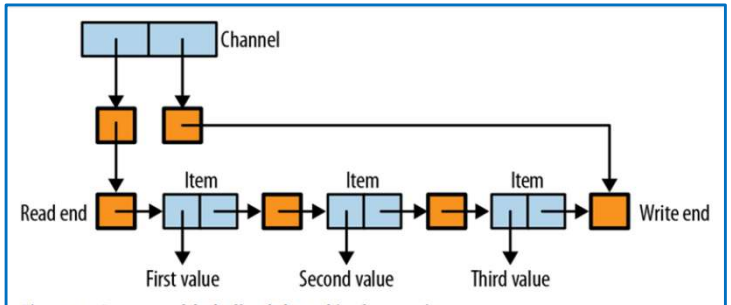


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
```

```
writeChan :: Chan a  -> a -> IO()
writeChan (Chan rV wV) val = do
            newStream <- newEmptyMVar
            writeEnd <- takeMVar wV
            putMVar writeEnd (Item val newStream)
            putMVar wV newStream
```

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Implementarea canalelor

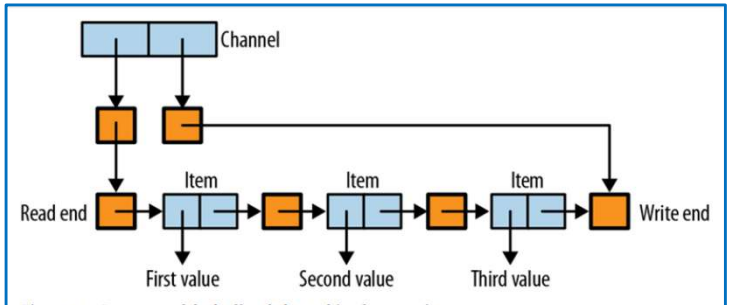


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
```

```
readChan :: Chan a -> IO a
readChan (Chan rV wV) = do
                stream <- takeMVar rV
                Item val str <- takeMVar stream
                putMVar rV  str
                return val
```

```
newChan :: IO(Chan a)
newChan = do
    emptyStream <-  newEmptyMVar
    readVar <- newMVar emptyStream
    writeVar <-newMVar emptyStream
    return (Chan readVar writeVar)
```

```
writeChan :: Chan a  -> a -> IO()
writeChan (Chan rV wV) val = do
          newStream <- newEmptyMVar
          writeEnd <- takeMVar wV
          putMVar writeEnd (Item val newStream)
          putMVar wV newStream
```

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Exercitiu: implementarea canalelor multicast

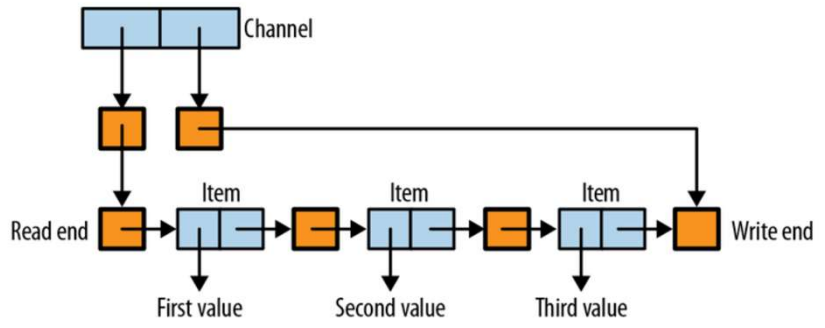


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
newChan :: IO (Chan a)
writeChan :: Chan a -> a -> IO ()
readChan :: Chan a -> IO a
```

dupChan :: Chan a -> IO (Chan a)

- noul canal este initial gol
- dupa crearea canalului duplicat, ceea ce se scrie pe oricare dintre canale poate fi citit de pe oricare dintre cele doua canale
- citirea de pe un canal **nu** elimina elementul de pe celalalt canal.

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Exercitiu: implementarea canalelor multicast

```
dupChan  ::  Chan a -> IO (Chan a)
```

- noul canal este initial gol
- dupa crearea canalului duplicat, ceea ce se scrie pe oricare dintre canale poate fi citit de pe oricare dintre cele doua canale
- citirea de pe un canal **nu** elimina elementul de pe celalalt canal.

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
newChan :: IO (Chan a)
writeChan :: Chan a -> a -> IO ()
readChan :: Chan a -> IO a
```

```
main = do  c <- newChan
           writeChan c 'a'
           readChan c >>= print
           dc <- dupChan c   -- creare canal duplicat
           writeChan c 'b'
           readChan c >>= print
           readChan dc >>= print
```

```
Prelude> :l chan2.hs
[1 of 1] Compiling Main
Ok, modules loaded: Main.
*Main> main
'a'
'b'
'b'
```

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Exercitiu: implementarea canalelor multicast

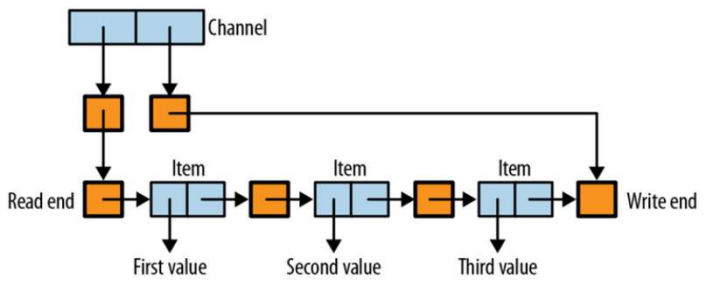


Figure 7-1. Structure of the buffered channel implementation

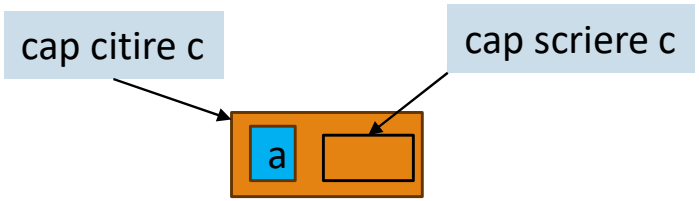
```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
newChan :: IO (Chan a)
writeChan :: Chan a -> a -> IO ()
readChan :: Chan a -> IO a
```

Canalul duplicat are **acelasi cap de scriere**, dar **un alt cap de citire**

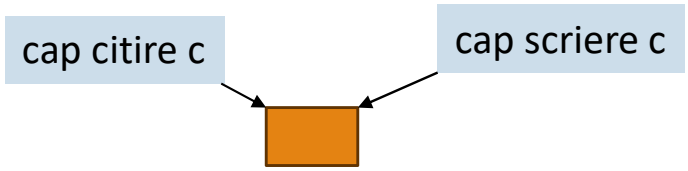
```
dupChan  ::  Chan a -> IO (Chan a)
dupChan (Chan _ wV) = do
                        writeEnd <- readMVar wV
                        newReadVar <- newMVar writeEnd
                        return (Chan newReadVar wV)
```

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

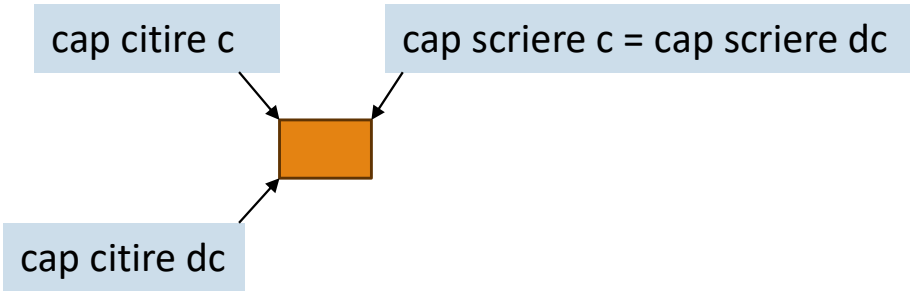
```
main = do
    c <- newChan
    writeChan c 'a'
```



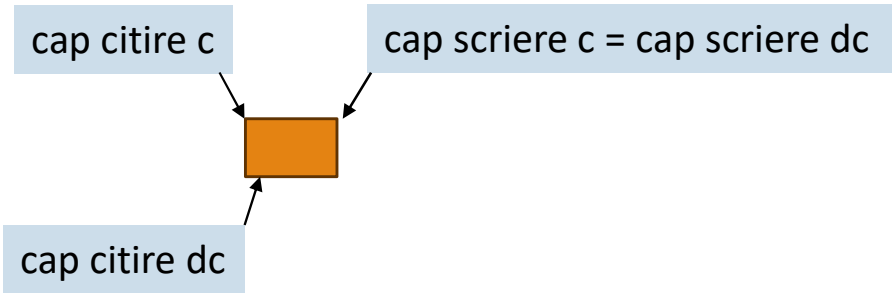

```
    readChan c >>= print
```



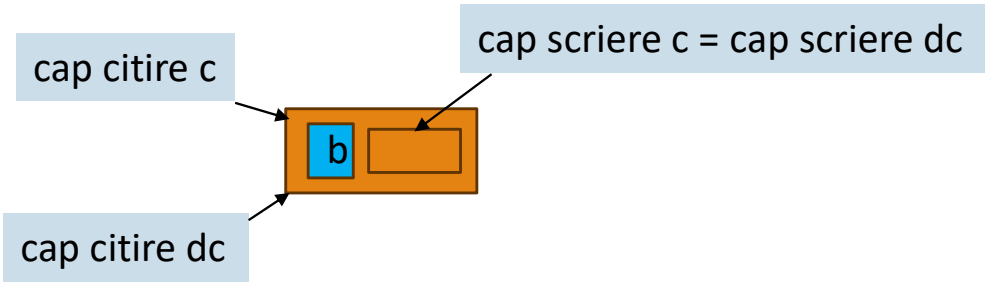

```
    dc <- dupChan c
```



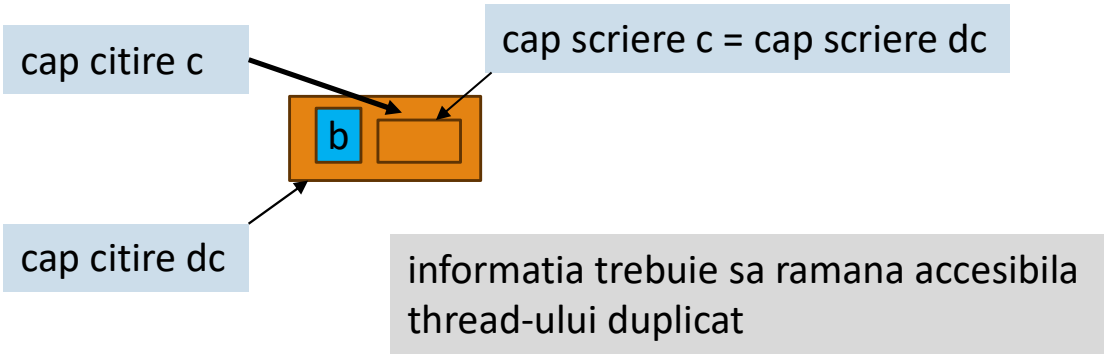
```
main = do
    ...
    dc <- dupChan c
```




```
    writeChan c 'b'
```




```
    readChan c >>= print
```




```
    readChan dc >>= print
```

➢ Exercitiu: implementarea canalelor multicast

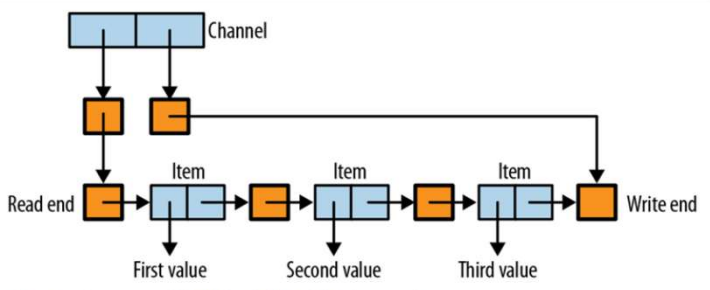


Figure 7-1. Structure of the buffered channel implementation

```
type Stream a = MVar (Item a)
data Item a   = Item a (Stream a)
data Chan a = Chan (MVar (Stream a)) (MVar (Stream a))
newChan :: IO (Chan a)
writeChan :: Chan a -> a -> IO ()
readChan :: Chan a -> IO a
```

```
dupChan  ::  Chan a -> IO (Chan a)
dupChan (Chan _ writeVar) = do
                        writeEnd <- readMVar writeVar
                        newReadVar <- newMVar writeEnd
                        return (Chan newReadVar writeVar)
```

operatia **readChan** trebuie modificata

```
readChan :: Chan a -> IO a
readChan (Chan rV wV) = do
                        stream  <- takeMVar rV
                        Item val str <- readMVar  stream
                        putMVar rV  str
                        return val
```

readMVar este folosit in locul lui **takeMVar** deoarece continutul trebuie sa ramana accesibil celuilalt canal.

http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_channels

➢ Comunicare sincrona

Un thread – reader- citeste un fisier text linie cu linie

Liniile sunt trimise, pe rand,
unui al doilea thread – writer- care le afiseaza
in ordinea trimisa si le numara.

La sfarsit, thread-ul writer trimite thread-ului reader
numarul de linii si acesta il afiseaza,

http://rosettacode.org/wiki/Synchronous_concurrency

```
import Control.Concurrent
import Control.Concurrent.MVar

main = do
    lineVar <- newEmptyMVar
    countVar <- newEmptyMVar
    forkIO $ writer lineVar countVar
    reader  lineVar countVar
```

- Comunicare sincrona

```
import Control.Concurrent
import Control.Concurrent.MVar

main = do
    lineVar <- newEmptyMVar
    countVar <- newEmptyMVar
    forkIO $ writer lineVar countVar
    reader  lineVar countVar
```

```
-- main thread
reader lineVar countVar  = do
        ls <- fmap lines (readFile "input.txt")
        mapM ((putMVar lineVar) . Just) ls
        putMVar lineVar Nothing
        n <- takeMVar countVar
        print n
```

```
readFile :: FilePath -> IO String
lines :: String -> [String]
fmap :: Functor f => (a -> b) -> f a -> f b
mapM :: Monad m => (a -> m b) -> [ a] -> m [ b]
```

http://rosettacode.org/wiki/Synchronous_concurrency

- Comunicare sincrona

```
import Control.Concurrent
import Control.Concurrent.MVar

main = do
    lineVar <- newEmptyMVar
    countVar <- newEmptyMVar
    forkIO $ writer lineVar countVar
    reader  lineVar countVar
```

```
readFile :: FilePath -> IO String
lines :: String -> [String]
fmap :: Functor f => (a -> b) -> f a -> f b
mapM :: Monad m => (a -> m b) -> [ a] -> m [ b]
```

```
reader lineVar countVar  = do
        ls <- fmap lines (readFile "input.txt")
        mapM ((putMVar lineVar) . Just) ls
        putMVar lineVar Nothing
        n <- takeMVar countVar
        print n

writer lineVar countVar = loop 0
    where
      loop n = do
                l <- takeMVar lineVar
                case l of
                  Just x  -> do putStrLn x
                                loop (n+1)
                  Nothing -> putMVar countVar n
```

http://rosettacode.org/wiki/Synchronous_concurrency

- Comunicare sincrona

```
import GetURL
import Data.ByteString as B

action x =  do
            r <- getURL x
            print (B.length r)
```

```
reader lineVar countVar  = do
          ls <- fmap lines (readFile "inputurl.txt")
          mapM ((putMVar lineVar) . Just) ls
          putMVar lineVar Nothing
          n <- takeMVar countVar
          print n

writer lineVar countVar = loop 0
     where
       loop n = do
                    l <- takeMVar lineVar
                    case l of
                       Just x  -> do
                                      action x
                                      loop (n+1)
                       Nothing -> putMVar countVar n
```

Fisierul "inputurl.txt" contine adrese web, iar **action** descarca paginile respective

https://hackage.haskell.org/package/parconc-examples-0.1/src/GetURL.hs

"**Concurrent computing** is a form of computing in which several computations are executing during overlapping time periods—*concurrently*—instead of *sequentially* (one completing before the next starts)[…]
A *concurrent system* is one where a computation can advance without waiting for all other computations to complete; where more than one computation can advance at *the same time*."
Operating System Concepts 9th edition, Abraham Silberschatz

Exemplu: incarcarea mai multor pagini web

secvential

concurent

Thread 1

Thread 2

```
import Data.ByteString as B
import GetURL
main = do
        m1 <- newEmptyMVar
        forkIO $ do
                r  <- getURL "http://..."
                putMVar m1 r

        m2 <- newEmptyMVar
        forkIO $ do
                r <- getURL "http://..."
                putMVar m2 r

         r1 <- takeMVar m1
         r2 <- takeMVar m2

     print (B.length r1, B.length r2)
```

geturl1.hs ©2012, Simon Marlow

- Comunicare asincrona

  Se creaza un thread separat pentru fiecare actiune si se asteapta rezultatul

```
var <- newEmptyMVar
forkIO $ do
            r  <- getURL "http://... "
            putMVar var  r
 takeMVar var
```

```
a <- async (getURL "http://... " )
wait a
```

```
data Async a = Async (MVar a)

async :: IO a -> IO (Async a)
async action = do
  var <- newEmptyMVar
  forkIO (do r <- action; putMVar var r)
  return (Async var)

wait :: Async a -> IO a
wait (Async var) = readMVar var
```

readMVar nu devine goala dupa citire (multiple-wakeup) deci mai multe apeluri **wait** pot fi facute pentru aceeasi operatie asincrona

- Comunicare asincrona

  Se creaza un thread separat pentru fiecare actiune si se asteapta rezultatul

```
data Async a = Async (MVar a)

async :: IO a -> IO (Async a)
async action = do
  var <- newEmptyMVar
  forkIO (do r <- action; putMVar var r)
  return (Async var)

wait :: Async a -> IO a
wait (Async var) = readMVar var
```

```
a <- async action
r <- wait a
```

```
import Data.ByteString as B
import GetURL

main = do
        a1 <-  async (getURL "http://..." )
        a2 <-  async (getURL "http://..." )
        r1 <- wait a1
        r2 <- wait a2
        print (B.length r1, B.length r2)
```

- Comunicare asincrona

```
data Async a = Async (MVar a)

async :: IO a -> IO (Async a)
async action = do
  var <- newEmptyMVar
  forkIO (do r <- action; putMVar var r)
  return (Async var)

wait :: Async a -> IO a
wait (Async var) = readMVar var
```

```
a <- async action
r <- wait a
```

```
import Data.ByteString as B
import GetURL
import TimeIt
import Text.Printf

timeDownload :: String -> IO ()
timeDownload url = do
        (page, time) <- timeit $ getURL url
        printf " %s (%d bytes, %.2fs)\n" url (B.length page) time


main = do
            a1 <-  async (timeDownload "http://..." )
            a2 <-  async (timeDownload "http://..." )
             wait a1
            wait a2
```

➢ Comunicare asincrona

```
import Data.ByteString as B
import GetURL
import TimeIt
import Text.Printf

timeDownload :: String -> IO ()
timeDownload url = do
        (page, time) <- timeit $ getURL url
        printf "downloaded: %s (%d bytes, %.2fs)\n" url (B.length page) time

main = do
            a1 <-  async (timeDownload "http://..." )
            a2 <-  async (timeDownload "http://..." )
            wait a1
            wait a2
```

vor fi afisate statistici privind timpul de executie

```
*Main> :set +s
*Main> main
downloaded: http://old.uefiscdi.ro/ (74599 bytes, 3.96)
downloaded: http://old.uefiscdi.ro/ (74599 bytes, 4.01)
(4.01 secs, 52,080 bytes)
```